Creating an LLVM Runtime

Michael Kruse *(michael.kruse@amd.com)*

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Outline

Introduction

Let’s Create a Runtime

Are We Done Yet?

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Outline

Introduction
- Why This Tutorial?
- History
- Rationale
- Goal

Let's Create a Runtime

Are We Done Yet?

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Why This Tutorial

[redacted] 1 hour ago Member

After this change, it's no longer possible to run openmp tests in a non-bootstrapping build (i.e. without building clang at the same time).

Previously it was possible to run the tests in a standalone build, by building the `FileCheck` and `not` binaries and pointing at them with `OPENMP_LLVM_TOOLS_DIR` - like I did at mstorsjo/llvm-mingw@ `20260210` /.github/workflows/build.yml#L583-L612.

Now after this change (or before, when doing a runtimes build), running tests only works if there's a `FileCheck` target already in the build, i.e. if doing a bootstrapping build:

llvm-project/openmp/cmake/OpenMPTesting.cmake
Lines 52 to 74 in c6425aa

```
52    if (${OPENMP_STANDALONE_BUILD})
53      find_standalone_test_dependencies()
54
55      # Set lit arguments.
56      set(DEFAULT_LIT_ARGS "-sv --show-unsupported --show-xfail")
57      if (MSVC OR XCODE)
58        set(DEFAULT_LIT_ARGS "${DEFAULT_LIT_ARGS} --no-progress-bar")
59      endif()
60      if ("${CMAKE_SYSTEM_NAME}" MATCHES "AIX")
61        set(DEFAULT_LIT_ARGS "${DEFAULT_LIT_ARGS} --time-tests --timeout=3000")
62      endif()
63      set(OPENMP_LIT_ARGS "${DEFAULT_LIT_ARGS}" CACHE STRING "Options for lit.")
```


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Why This Tutorial

[redacted]

[redacted] on Dec 8, 2025 • edited

Yep, just for context from the Rust side. We have a lot of builders in CI, but the slowest one we have is the x86 dist builder, which we use to e.g. create compiler artifacts. We already build LLVM there around 4 times due to PGO and Bolt, so we really try hard not to make it any slower, since we are already at ~3.5 hrs, and we run this CI before any merge into main.

One of the compromises there is that we have two checkouts of LLVM - once as a standalone llvm/llvm-project build (A) and once as a submodule of rustc in `src/tools/llvm-project`. We use build A to compile C/C++ code, and keep rebuilding the second checkout (B) during bootstrapping.
We only build clang as part of our LLVM Build A. However, rustc is based on LLVM B, and as such we'd also want the openmp and offload libraries to also be part of our LLVM Build B, which does not build the in-tree clang to save compile times.

We've tried for quite a bit to get the runtimes to build despite no in-tree clang or lld in LLVM B, and failed, I'm happy to share all the issues I ran into there. A standalone build instead allows me to use clang from build A to build offload+openmp from checkout B, that's compatible with what we have in rustc. So until that is fixed, standalone builds are likely the only builds that our infra team would be ok with.

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Why This Tutorial

[redacted]

[redacted] on Dec 8, 2025 • edited

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[redacted] 1 hour ago • edited Contributor

We were using the OpenMP standalone build mode in order to build the static libomp.a runtime library which cannot be build together with the shared libomp.so (there is no FLANG_RT_ENABLE_SHARED and FLANG_RT_ENABLE_STATIC equivalent for libomp, the LIBOMP_ENABLE_SHARED flag is binary: one can get either shared or static library, but never both in the same build).

The OpenMP standalone build mode has been removed by llvm/llvm-project#182022

This complicates our situation. Considering almost nonexistent demand for the static OpenMP runtime library, it is not worth pursuing it.

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

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[redacted] on Jan 7 • edited

Member

That comes with two drawbacks: 1) checking out `llvm` alone (I'm not talking about `llvm-project`) takes a lot of time. 2) the way to pass CMake arguments to runtime build is extremely user unfriendly. If these two can't be improved, I strongly object the removal of standalone build.

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build t
for libc
build).

The Op

This co... pleted our situation. Considering almost nonexistent demand for the static OpenMP runtime library, it is not worth pursuing it.

om

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

Yep, just for context from the R...

[redacted] on Dec 8, 2025 Member

Rust colleagues rely on standalone builds for their offloading extensions (see rust-lang/rust#148671).

CC @ZuseZ4

the removal of standalone build.

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

Yep, just for context from the [redacted]

[redacted] on Jul 26, 2025

Member

It is definitely not the same!
For standalone build, I used to not needing the llvm-project tree. I could build libomp by just downloading the OpenMP release tarball: `llvmorg-21.1.0-rc1` openmp-21.1.0-rc1.src.tar.xz (download)

I just realized that this workflow got bricked by #62240

[redacted] complicates our situation. Considering almost nonexistent demand for the static OpenMP runtime library, it is not worth pursuing it.

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

Yep, just for context from the [redacted]

Member

[redacted] on Jan 7

Member

I don't have a build tree. That's the whole point of building standalone. And the installed binary tree doesn't work.

I just realized that this w[redacted]

[redacted] almost nonexistent demand for the static Openm[redacted]

pursuing it.

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History

1. Initially, each subproject had its own SVN repository
   - Subprojects categorized as “tools” (clang: 2009), “runtime” (libprofile), and “projects” (e.g. llvm-gcc)
   - An example project boilerplate existed

[1] https://reviews.llvm.org/D20992
[2] https://lists.llvm.org/pipermail/llvm-dev/2020-October/145997.html
[3] https://reviews.llvm.org/D26365
[4] https://reviews.llvm.org/D132478

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2. “runtime” subdirectory disappears with dysfunct PGO removal (2013)

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3. Desire to separate runtimes[1][2] (2016–2020)
   - Somewhat different from compile-time utils/tools/compilers

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3. Desire to separate runtimes[1,2] (2016–2020)
   - Somewhat different from compile-time utils/tools/compilers
4. Introduction of `LLVM_ENABLE_PROJECTS`[3] (2016)
5. Introduction of `LLVM_ENABLE_RUNTIMES`[4] (2022)
   - First for libc++

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Rationale

Why `LLVM_ENABLE_RUNTIMES`?

Build Host

The system we build on

- `cmake`, `ninja`, `cc`, ...
- `llvm-tblgen`, `mlir-tblgen`, ...

Cross-Compile Target

Where Clang will run

- `clang`, `opt`, `lld`, ...
- `LLVMSupport{.a/.so}`, `LLVMPolly.so`, ...

Compiler Target

What Clang will compile for

- `a.out`
- compiler-rt, `libc`, `libc++`, ...

- Automate multiple build-steps
  - Instead of: Build clang → build compiler-rt for x86_64 → build compiler-rt for arm64 → ...
- Reduce per-runtime boilerplate/differences between build modes
- Runtimes compiled (Stage 2) using just-built Clang

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Rationale

Why `LLVM_ENABLE_RUNTIMES`?

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- `cmake`, `ninja`, `cc`, ...
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- `clang`, `opt`, `lld`, ...
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What Clang will compile for

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Stage 1

Stage 2

- Automate multiple build-steps
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- Reduce per-runtime boilerplate/differences between build modes
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Tutorial Runtime

What is our tutorial runtime going to do?

Our Runtime

```
void _dublin_hello_world() {
  puts("Dia daoibh, a dhomhain!\n");
}
```

- Use LLVM best-practices
- Linux only
- Prioritize conciseness over
  - … features
  - … comments
  - … boilerplate not topic in this tutorial
- Point out shortcomings of the current system

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Outline

Introduction

Let's Create a Runtime

- Setup
- Step 1: Register with the LLVM Build System
- Step 2: Build a Library Artifact
- Step 3: An Example that uses the Library
- Step 4: Build Modes
- Step 5: CMake Cache Files
- Step 6: Artifact Output Location
- Step 7: Installation
- Step 8: Shared and Static Libraries
- Step 9: Regression Testing
- Step 10: Unittests
- Step 11: Sphinx Docs
- Step 12: Doxygen Docs
- Step 13: Cross-Compilation
- Step 14: Accelerator Offloading
- Step 15: Depending on Other LLVM Libraries / C++

Are We Done Yet?

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Setup

https://github.com/Meinersbur/llvm-dublin/tree/dublin-0-main

- README.md
  - Update Readme for tutorial
- .gitignore
  - Make git ignore `install*` subdirs (like `build*` already is)

$HOME/bin/cmake

```
/usr/bin/cmake -G Ninja                                  \
               -D CMAKE_C_COMPILER_LAUNCHER=ccache       \
               -D CMAKE_CXX_COMPILER_LAUNCHER=ccache     \
               -D CMAKE_DISABLE_PRECOMPILE_HEADERS=ON    \
               "$@"
```


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Register with the LLVM Build System

https://github.com/Meinersbur/llvm-dublin/tree/dublin-1-basic

- Allow `LLVM_ENABLE_RUNTIME=dublin`
  - `llvm/CMakeLists.txt`
  - `runtimes/CMakeLists.txt`
- Subproject boilerplate
  - `dublin/CMakeLists.txt`
  - `dublin/CODE_OWNERS.TXT`
  - `dublin/LICENSE.TXT`

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Step 2: Build a Library Artifact

- Library source files
  - `lib/Dublin/dublin.c`
  - `include/dublin/Dublin/dublin.h`
- Build `libdublin.a`
  - `add_executable(dublin dublin.c)`

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Standard Subproject Layout

```
subproject
├─ docs
│  └─ *.rst, *.md
├─ examples
│  └─ Example1
│     └─ *.c, *.cpp
├─ test
│  ├─ Unit
│  │  └─ lit.cfg.py, lit.site.cfg.py.in
│  ├─ Component1
│  │  └─ *
│  ├─ Component2
│  │  └─ *
│  └─ lit.cfg.py, lit.site.cfg.py.in
└─ unittests
   ├─ Component1
   │  └─ Component1Test.cpp
   └─ Component2
      └─ Component1Test.cpp
```

```
├─ cmake
│  ├─ Modules/modules
│  │  └─ *.cmake
│  └─ caches
│     └─ *.cmake
├─ include
│  └─ subproject
│     ├─ Component1
│     │  └─ component1.h
│     └─ Component2
│        └─ component2.h
├─ lib
│  ├─ Component1
│  │  ├─ component1.c
│  │  └─ component1-internal.h
│  └─ Component2
│     └─ component2.c
└─ utils
   └─ utility
      └─ utility.cpp
```


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An Example that uses the Library

`https://github.com/Meinersbur/llvm-dublin/tree/dublin-3-example`

- Example source
  - `examples/dublin-hello/dublin-hello.c`
- Build instructions
  - `examples/dublin-hello/CMakeLists.txt`

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Build Modes

- Detecting Legacy Build Modes
  - `CMakeLists.txt`

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Build Modes

Legacy Build Modes

As project: `cmake -S <llvm-project>/llvm -D LLVM_ENABLE_PROJECTS=dublin`
Standalone: `cmake -S <llvm-project>/dublin`

- Intended to be superseded by runtimes build modes[1]
- If your build infrastructure still uses this, please consider updating

Runtimes Build Modes

Bootstrapping: `cmake -S <llvm-project>/llvm -D LLVM_ENABLE_RUNTIMES=dublin`
Default/Standalone: `cmake -S <llvm-project>/runtimes -D LLVM_ENABLE_RUNTIMES=dublin`

- Shared+common boilerplate code in `<llvm-project>/runtimes/CMakeLists.txt`

[1] https://lists.llvm.org/pipermail/llvm-dev/2020-October/145997.html

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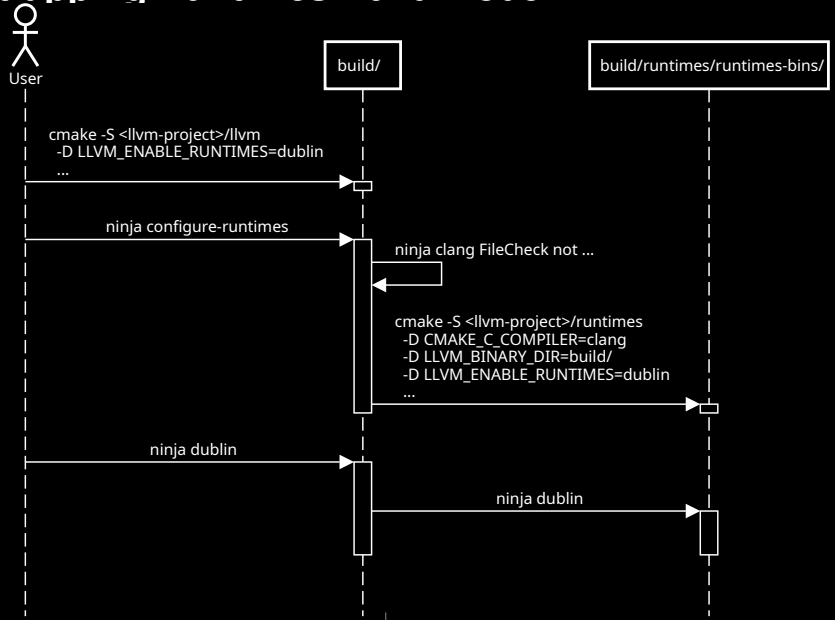
Bootstrapping Runtimes Build Mode



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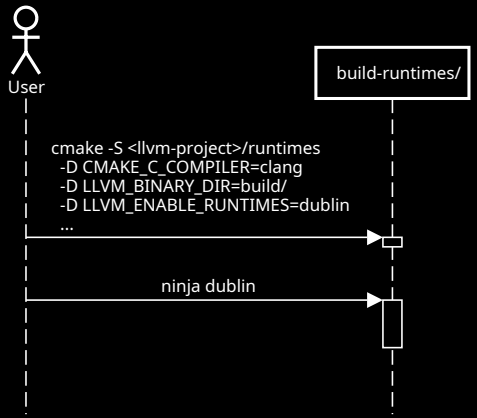
Default/Standalone Runtimes Build Mode

Skip the Middleman



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Default/Standalone Runtimes Build Mode

runtimes/CMakeLists.txt

```
project(Runtimes C CXX ASM)

find_package(LLVM HINT ${LLVM_BINARY_DIR})
# more boilerplate code

foreach (runtime ${LLVM_ENABLE_RUNTIMES})
  add_subdirectory(${runtime})
endforeach ()
```


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Bootstrapping Build Mode

`llvm/runtimes/CMakeLists.txt`

- Configure nested builddirs
  - `runtimes/buitins-bins` for Compiler-RT
  - `runtimes/runtimes-bins`
- Forward configuration parameters
  - Selected configuration options (e.g. `LLVM_INCLUDE_TESTS`)
  - Any option that starts with the capitalized runtime name (i.e. `DUBLIN_*`)
  - The content of `RUNTIMES_CMAKE_ARGS`
    e.g. `-DRUNTIMES_CMAKE_ARGS="-DCMAKE_CXX_FLAGS=-ferror-limit=1"`
- Build prerequisites
  - Test infrastructure: `llvm-lit`, `FileCheck`, `not`, ...
  - Compilers: Clang, Flang (if enabled via `CMAKE_ENABLE_PROJECTS`)
- Build targets for the nested builddir(s)
  - Control targets: `configure-runtimes`, `install-runtimes`, `clean-runtimes`, ...
  - Added dependency: `all`, `install`
  - Just forwarded: `dublin`, `check-dublin`, `runtimes` → `all`

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CMake Cache Files

`https://github.com/Meinersbur/llvm-dublin/tree/dublin-5-cachefiles`

- `cmake/caches/runtimes-bins.cmake`
  - Pre-populated CMake parameters as a bootstrapping build would set them

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Artifact Output Location

https://github.com/Meinersbur/llvm-dublin/tree/dublin-6-location

- `CMakeLists.txt`, `cmake/Modules/GetToolchainDirs.cmake`
  - Determine standard locations
- `lib/Dublin/CMakeLists.txt`
  - Set artifact output location

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Library Locations

`LLVM_ENABLE_PER_TARGET_RUNTIME_DIR=ON`

Prefix Dir

```
$PREFIX
├─ bin
│  └─ clang
│
└─ lib
   └─ <triple>
      └─ *.so, *.a
```

- Accessible to any compiler

Resource Dir

```
$PREFIX
├─ bin
│  └─ clang
│
└─ lib
   └─ clang
      └─ <version>
         └─ lib
            └─ <triple>
               └─ *.a
```

- Accessible only for this version of Clang (and Flang)

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Installation

`https://github.com/Meinersbur/llvm-dublin/tree/dublin-7-install`

- `CMakeLists.txt`
  - Set install output location
- `lib/Dublin/CMakeLists.txt`
  - Library installation
- `include/CMakeLists.txt`
  - Header installation

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Shared and Static Libraries

`https://github.com/Meinersbur/llvm-dublin/tree/dublin-8-shared`

- `cmake/Modules/AddDublin.cmake`
  - If building Shared and Static library, use an OBJECT library
- `lib/Dublin/CMakeLists.txt`
  - Use `add_dublin_library` from `AddDublin.cmake`

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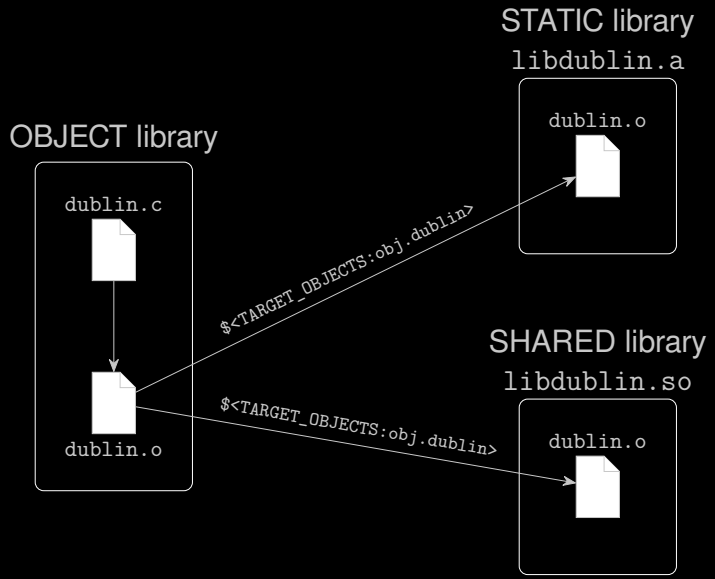
OBJECT Libraries



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Regression Testing

https://github.com/Meinersbur/llvm-dublin/tree/dublin-9-test

- dublin/test/Dublin/dublin-hello.c
  - The regression test:
    - Build the source
    - Link to libdublin(.a/.so)
    - Run
    - Expect output: "Dia daoibh, a dhomhain!"
- dublin/test/CMakeLists.txt
  - Testing boilerplate
- dublin/test/lit.cfg.py, dublin/test/lit.site.cfg.py.in,
  - llvm-lit configuration
    - How to find regression tests (lit.formats.ShTest)
    - Setup execution environment
    - Forward build configuration

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Unittests

`https://github.com/Meinersbur/llvm-dublin/tree/dublin-10-unittest`

- `unittests/Dublin/DublinTest.cpp`
  - Google Test-based test code
- `test/Unit/lit.cfg.py`, `test/Unit/lit.site.cfg.py.in`
  - `llvm-lit` configuration
    - How to find unittests (`lit.formats.GoogleTest`)

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Sphinx Docs

https://github.com/Meinersbur/llvm-dublin/tree/dublin-11-docs

- docs/index.md
  - The documentation
- docs/conf.py, docs/CMakeLists.txt
  - Sphinx configuration
- docs/_theme
  - HTML boilerplate

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Doxygen Docs

https://github.com/Meinersbur/llvm-dublin/tree/dublin-12-doxygen

- `lib/Dublin/dublin.c`
  - Add Doxygen comment
- `docs/doxygen-mainpage.dox`
  - Doxygen frontpage content
- `docs/doxygen.cfg.in`, `docs/CMakeLists.txt`
  - Doxygen configuration

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Cross-Compilation

https://github.com/Meinersbur/llvm-dublin/tree/dublin-13-cross

No modifications necessary

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Cross-Compilation

- `-D LLVM_RUNTIME_TARGETS="default;aarch64-linux-gnu"`
  Compile runtimes also for arm64
- `-D RUNTIMES_CMAKE_ARGS=-DLLVM_USE_LINKER=lld`
  Ubuntu's default `ld.bfd` does not support arm64
- `-D RUNTIMES_aarch64-linux-gnu_ZLIB_LIBRARY=/usr/lib/aarch64-linux-gnu/libz.a`
  CMake `find_package(ZLIB)` may not find multilib libraries

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Accelerator Offloading

https://github.com/Meinersbur/llvm-dublin/tree/dublin-14-offload

- `CMakeLists.txt`, `cmake/Modules/AddDublin.cmake`
  - Accommodate imitations of GPGPU targets
- `examples/dublin-offload`
  - Demonstrate OpenMP offloading

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Depending on Other LLVM Libraries / C++

`https://github.com/Meinersbur/llvm-dublin/tree/dublin-15-deps`

- `lib/Dublin/dublin.cpp`, `include/dublin/Dublin/dublin.h`, `examples/dublin-hello/dublin-hello.cpp`
  - Convert to C++
- `cmake/Modules/AddDublin.cmake`, `third-party/unittest/CMakeLists.txt`
  - Link to libc++, regardless of compiler default

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Outline

Introduction

Let's Create a Runtime

Are We Done Yet?

- More Topics
- Build System Improvements

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Not Covered Topics

- Platform Introspection
  `check_source_compiles`
- Platform Support
  Windows/MacOS/AIX/Fuchsia/…
- Multilib
  `lib32/lib64`
- Export libraries
  `find_package(Dublin)`
- Continuous Integration
- Fortran
- …

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Potential Runtimes Build Improvements

- Inform LLVM developers and OS distributions

  (Build modes) developer documentation missing, runtime template

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Potential Runtimes Build Improvements

- Inform LLVM developers and OS distributions

  (Build modes) developer documentation missing, runtime template
- Make `RUNTIMES_*` options discoverable

```
set(RUNTIMES_CMAKE_ARGS "" CACHE STRING "Help")
```


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  `set(RUNTIMES_CMAKE_ARGS "" CACHE STRING "Help")`
- Unify the CMake code of all runtimes

  Boilerplate diverged significantly since the SVN days

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  MacOS Universal Binaries, `arm64ec`, Offload, …

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- Reduce special cases
  - GPGPU targets
  - `LLVM_RUNTIME_TARGETS=default`
  - Compiler-RT “builtins”

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- Inform LLVM developers and OS distributions

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- Make `RUNTIMES_*` options discoverable

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  MacOS Universal Binaries, `arm64ec`, Offload, …
- Reduce special cases
  - GPGPU targets
  - `LLVM_RUNTIME_TARGETS=default`
  - Compiler-RT “builtins”
- Writing runtimes in C++

  Option whether to use default stdlib or libc++

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Potential Runtimes Build Improvements

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  MacOS Universal Binaries, `arm64ec`, Offload, …
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  - GPGPU targets
  - `LLVM_RUNTIME_TARGETS=default`
  - Compiler-RT “builtins”
- Writing runtimes in C++

  Option whether to use default stdlib or libc++
- Get rid of `LLVM_ENABLE_PER_TARGET_RUNTIME_DIR`

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Potential Runtimes Build Improvements

- Inform LLVM developers and OS distributions

  (Build modes) developer documentation missing, runtime template
- Make `RUNTIMES_*` options discoverable

  `set(RUNTIMES_CMAKE_ARGS "" CACHE STRING "Help")`
- Unify the CMake code of all runtimes

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- QEMU (`CMAKE_CROSSCOMPILING_EMULATOR`)
- …

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